Chapter 27
Counterparty Credit Risk and Capital

Chapter 9 priced counterparty risk. This chapter capitalizes it, and the two are not the same activity.

CVA is a price: what the risk is worth, computed with risk-neutral probabilities and hedgeable in principle. Regulatory capital is a buffer: what must be held against the possibility that the price was wrong. *Capital in this chapter always means regulatory capital*: equity and other loss-absorbing instruments a bank is required to hold, set by the Basel framework of Section 9.7 and written into national rules by each jurisdiction's regulator, which is why the same standard binds differently in Europe, the United States, and China. Section 9.7 flagged that a firm can reduce one without reducing the other, and the reason is that they are computed by different rules for different purposes: one by the firm's own model against market prices, the other by a standardized formula against a supervisory scenario.

The chapter runs from the exposure to the framework that measures it, and then to what the framework misses. Sections 27.1 and 27.2 are the object itself and the legal clause that reduces it most. Sections 27.3 to 27.5 are the standardized approach and the capital it feeds, and Sections 27.6 to 27.9 the charges around it, including wrong-way risk. Sections 27.10 to 27.12 are the leverage backstop and a bank's whole requirement. Sections 27.13 to 27.17 are the limit: what a bilateral framework cannot see, and who bears it.

Section 7.10 said residual counterparty risk is capitalized under this framework and a 2021 episode showed how quickly it returns; Section 7.12 said total return swaps are the instrument through which leverage accumulates quietly and pointed here. Section 8.7 deferred wrong-way risk in general. Section 27.14 is where all three arrive together.

27.1 WHY IT NEEDS ITS OWN FRAMEWORK

A loan's exposure is known: it is the amount lent. A derivative's is not.

The exposure is an option on the market. A swap that is worth nothing today may be worth $50 million in two years, and the counterparty's failure matters only in the states where it is worth something to you. Section 9.1 made this point and Figure 9.1 drew the profile; the consequence for capital is that the amount at risk must be modeled rather than looked up.

Three features follow, and each requires machinery a loan book does not need.

It is bilateral. Either party may end up owing the other, so exposure can change sign, and both parties hold credit risk on the same contract at different times.

It is portfolio-dependent. The exposure to a counterparty is a property of the whole netting set, not of any trade in it. A new trade can reduce total exposure, which no loan ever does.

And it is mitigated by mechanisms that can themselves fail. Netting depends on enforceability, collateral on operational performance and on the value of what is posted, which Chapters 1 and 8 addressed and which the capital framework largely assumes will work.

Figure 27.1 draws that option over the life of a swap, as the expected exposure and the

95th-percentile potential future exposure (PFE) that Section 9.1 defined. Section 9.1 explained the hump: per unit of notional the PFE is about $(T - t) \times 1.645\,\sigma\sqrt{t}$ at a normal rate volatility σ, and the expected exposure is the same product with 0.40 in place of 1.645, so diffusion and the shrinking number of payments peak against each other about a third of the way through. The lowest curve is the same swap under daily variation margin with a ten-day margin period of risk. Margin settles the mark-to-market each day, so the only move that can accumulate is the one over the ten days it takes to close out a defaulted counterparty: the curve replaces $\sqrt{t}$ by $\sqrt{10/250}$, which is why it has no hump and simply declines with the payments left. It is also positive even at inception, when the swap is worth nothing and no margin is posted: a counterparty that fails on the first day still leaves ten days of moves uncovered. [TN27] derives all three curves. The gap between the margined curve and the others is what the maturity factor of Section 27.3 is about.

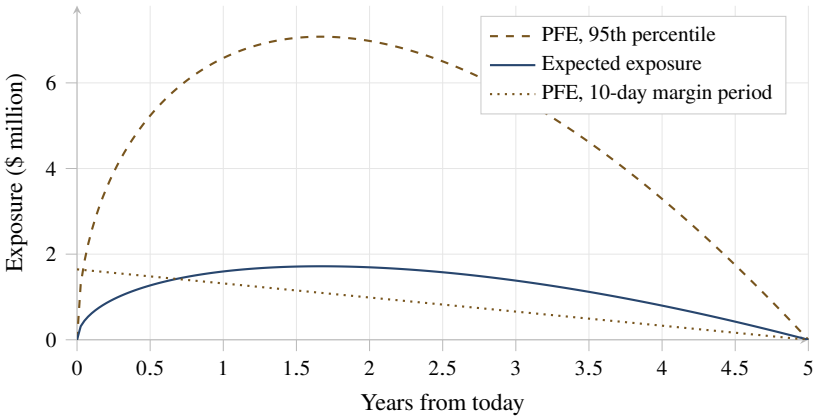


Figure 27.1: Exposure on a five-year par swap, $100 million notional, at a normal rate volatility of 100 basis points a year. Illustrative, not market data.

27.2 WHAT NETTING IS WORTH

Before any formula, Example 27.1 measures the single largest exposure reducer in the framework.

Example 27.1 The netting benefit

Five trades with one counterparty, current mark-to-market values in millions:

$$+40, \quad -25, \quad +15, \quad -30, \quad +10$$

Without an enforceable netting agreement, the exposure is the sum of the positive values: the counterparty's administrator would enforce the trades you owe on and pay only a recovery fraction on the trades owed to you. That is Chapter 1's cherry-picking, and it gives

$$40 + 15 + 10 = \$65 \text{ million}$$

With netting, the exposure is the net value:

$$\max(40 - 25 + 15 - 30 + 10,\ 0) = \$10 \text{ million}$$

An 85% reduction, from a legal clause rather than a hedge—and that is only today's exposure. The formula of Section 27.3 adds a second component for how much worse it could become before a default is closed out, and netting reduces that too, because offsetting trades partly cancel before it is computed.

This is why Section 1.5 called the enforceability of close-out netting a precondition for a derivatives market existing at all. The whole framework assumes netting is enforceable, and where

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it is not (no close-out netting statute, or a counterparty type carved out), exposure and capital revert to the gross figure. That is why a jurisdiction's **netting opinion** (the legal opinion, commissioned by the industry for each jurisdiction, that close-out netting will be upheld in a counterparty's insolvency there) is worth more to a dealer than any risk model.

27.3 SA-CCR, AND WHAT MARGINING BUYS

The **standardized approach for counterparty credit risk** (SA-CCR) computes exposure at default (EAD), the amount the capital rules treat as lent to the counterparty, for each netting set (Chapter 8), the trades with one counterparty that a single enforceable netting agreement covers:

$$\text{EAD} = \alpha \times (\text{RC} + \text{PFE}), \qquad \alpha = 1.4 \tag{27.1}$$

The two components are computed as follows; the scalar α is Section 27.5's subject.

Replacement cost. RC is what the counterparty's default would cost today. Write V for the current market value of the netting set to the bank, the sum of its trades' values, and C for the net collateral the bank holds against it after haircuts. Without a margin agreement,

$$\text{RC} = \max(V - C,\ 0) \tag{27.2}$$

With one, RC is the largest exposure that would *not* trigger a margin call:

$$\text{RC} = \max(V - C,\ \text{TH} + \text{MTA} - \text{NICA},\ 0) \tag{27.3}$$

where TH is the threshold below which no variation margin is called, MTA the minimum transfer amount, and NICA the net independent collateral amount (initial margin received less initial margin posted). The middle term is there because a margin agreement does not hold the exposure at zero: no call is made below the threshold, and then only above the minimum transfer, so the exposure can sit at TH + MTA with the counterparty in full compliance, and only initial margin in hand offsets that (Figure 27.2, left). With zero threshold and minimum transfer amount and no initial margin, the term vanishes and RC is the uncollateralized mark-to-market as before.

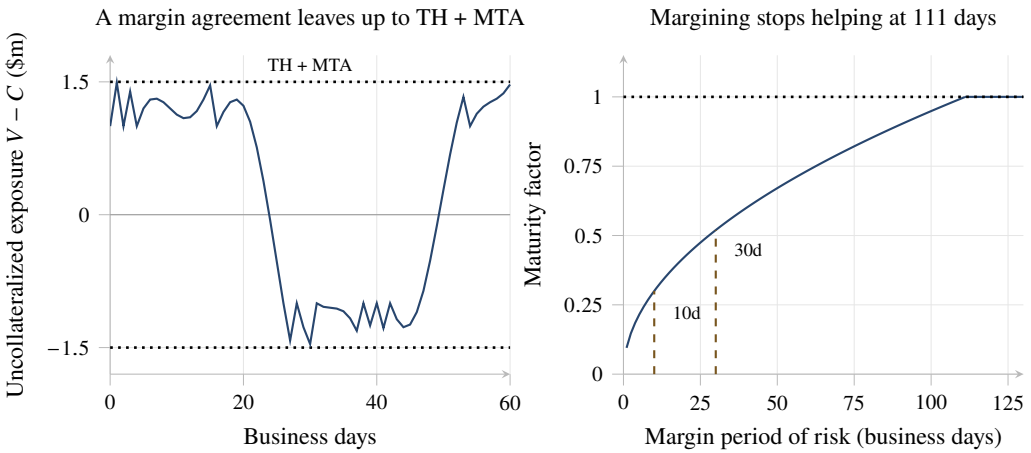


Figure 27.2: Left: uncollateralized exposure under a margin agreement (TH = \$1m, MTA = \$0.5m), illustrative path; each drop is a margin call. Right: equation (27.7), capped at one.

Potential future exposure. PFE is a supervisory estimate of how much worse the exposure

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could become before a default is closed out:

$$\text{PFE} = \text{multiplier} \times \sum_{\text{asset classes}} \text{AddOn} \tag{27.4}$$

The **add-on** for an asset class stands in for how far that class's positions can move over the horizon that matters. Within interest rates, the class Example 27.2 uses, trades are grouped into **hedging sets**, one per currency, and within a hedging set into three maturity buckets: under one year, one to five years, over five years. Each trade i contributes an **effective notional**

$$D_i = \delta_i \times d_i \times \text{MF}_i \times N_i \tag{27.5}$$

where N_i is the notional; δ_i is the **supervisory delta**, $+1$ for a position that gains when rates rise (paying fixed), -1 for one that loses, or a Black–Scholes delta for an option; d_i is the **supervisory duration**, the present value at a fixed 5% of one unit a year from the trade's start date t_i to its maturity T_i,

$$d_i = \frac{e^{-0.05 t_i} - e^{-0.05 T_i}}{0.05} \tag{27.6}$$

and MF_i is the **maturity factor**, which scales the position to the horizon over which it is exposed:

$$\text{MF} = \sqrt{\min(M, 1)} \ \text{ unmargined}, \qquad \text{MF} = 1.5\sqrt{\text{MPOR}/250} \ \text{ margined} \tag{27.7}$$

with M the remaining maturity in years and MPOR the margin period of risk in business days, the time assumed to pass between the last margin exchange and close-out: ten days for a daily bilateral agreement, five for client-cleared trades, twenty above 5,000 trades or with illiquid collateral, and double after repeated margin disputes. The 250 is the framework's business days in a year, so MPOR/250 is that window in years.

The product is not a notional in any economic sense. The first three factors are the trade's sensitivity to the rate ($\delta_i d_i N_i$ is its signed annuity, so Example 27.2's $D = 442.40$ million says the swap gains or loses \$44,240 per basis point), and MF_i carries the square root of the horizon over which that sensitivity is exposed. The volatility that turns the two into money sits in the supervisory factor below. So D_i is the amount of a one-year reference trade carrying the same exposure, which is what *effective* means in the name.

Effective notionals for the three maturity buckets $k = 1, 2, 3$ are summed within each, $D_k = \sum_{i \in k} D_i$, so offsetting trades cancel, and the buckets then combine with partial offset into the hedging set's effective notional EN:

$$\text{EN} = \sqrt{D_1^2 + D_2^2 + D_3^2 + 2 \times 0.7\, D_1 D_2 + 2 \times 0.7\, D_2 D_3 + 2 \times 0.3\, D_1 D_3} \tag{27.8}$$

Equation (27.8) is the standard deviation of a sum of correlated normal variables, with correlations of 0.7 between adjacent buckets and 0.3 between the outer two, and those correlations are what *partial* means. Take a payer of $D_2 = +100$ in the one-to-five-year bucket against a receiver of $D_3 = -100$ in the over-five-year bucket. Then (27.8) returns $\sqrt{100^2 + 100^2 - 2 \times 0.7 \times 100 \times 100} = 77$, not the 0 of a perfect hedge or the 141 of none: the position counts as mostly, not wholly, offset. Within a bucket the offset is complete, because there the D_i are added before anything is squared.

The hedging set's add-on is then

$$\text{AddOn}_{\text{hedging set}} = \text{SF} \times \text{EN} \tag{27.9}$$

with the **supervisory factor** SF = 0.50% for interest rates, and the asset class's add-on is the sum over its hedging sets. The other classes follow the same pattern with their own factors and hedging sets (4% per currency pair for foreign exchange, 32% for a single stock with partial

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offset across issuers for equity), and the Basel standard has the details. The approach covers derivatives of every kind, cleared as well as bilateral (which is what lets Example 27.3 put three regimes in one table, since clearing changes the risk weight and not the measure), but not securities financing.

The **multiplier** in (27.4) recognizes collateral in excess of the current exposure, or a netting set that is out of the money, both of which cushion future moves:

$$\text{multiplier} = \min\left\{1,\ 0.05 + 0.95\,\exp\left(\frac{V - C}{2 \times 0.95 \times \sum \text{AddOn}}\right)\right\} \tag{27.10}$$

It equals one whenever $V \geq C$, so for a netting set at or in the money the PFE is simply the sum of the add-ons, and it falls toward a floor of 5% as $C - V$ grows relative to the add-on.

None of this is arbitrary. The formula is a closed-form approximation of the internal-model quantity of Section 27.5: underneath it the netting set's value is normal with no drift, as in the model behind Figure 27.1, and the target is the effective expected positive exposure over one year (an average of the expected exposure over that year, defined precisely in Section 27.5), which α scales as it scales a bank's own model. Each piece corresponds:

- the supervisory factor is the one-year expected positive exposure of a unit at-the-money position at a stressed *normal* volatility, a move in rate rather than a percentage return (0.50% for rates corresponds to about 190 basis points a year);
- the maturity factor is the average of $\sqrt{t}$ over the horizon, discussed below;
- the bucket correlations behind (27.8) are a supervisory judgment that maturities move together but not identically;
- the multiplier is a floored exponential fitted to the exact normal expectation for a position that starts out of the money.

[TN27] works each correspondence through.

Example 27.2 A single swap, and what margin does to it

A five-year interest rate swap, \$100 million notional, in which the bank pays fixed. It is currently at zero mark-to-market, with no collateral and no margin agreement.

Replacement cost. $V = 0$ and $C = 0$, so by (27.2) RC = 0.

Effective notional. Paying fixed gains when rates rise, so $\delta = +1$. The trade starts now and runs five years, so $t = 0$ and $T = 5$, and by (27.6)

$$d = \frac{1 - e^{-0.25}}{0.05} = \frac{1 - 0.7788}{0.05} = 4.4240$$

Unmargined with five years remaining, $\text{MF} = \sqrt{\min(5, 1)} = 1$, so $D = 1 \times 4.4240 \times 1 \times 100 =$ \$442.40 million. It is the only trade in its hedging set and sits in the one-to-five-year bucket, so EN = 442.40.

Add-on, PFE, and EAD. AddOn = $0.50\% \times 442.40 =$ \$2.212 million. With $V - C = 0$ the multiplier is one, so PFE = 2.212 and

$$\text{EAD} = 1.4 \times (0 + 2.212) = \$3.10 \text{ million}$$

In the money. If the swap were instead worth \$5 million to the bank, RC = 5, the multiplier stays at one, and $\text{EAD} = 1.4 \times (5 + 2.212) =$ \$10.10 million.

Out of the money. If it were worth −\$5 million, RC = 0 and the multiplier does its work:

$$\text{multiplier} = 0.05 + 0.95\,e^{-5/(2\times0.95\times2.212)} = 0.05 + 0.95 \times 0.3043 = 0.339$$

so $\text{PFE} = 0.339 \times 2.212 =$ \$0.75 million and $\text{EAD} = 1.4 \times 0.75 =$ \$1.05 million. A swap the

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bank owes money on still carries exposure, because rates can move more than $5 million in its favor, but a third of the at-the-money amount.

Now add a margin agreement with daily exchange, zero threshold and minimum transfer amount, and a ten-day margin period of risk. At zero mark-to-market (27.3) still gives RC = 0, and the maturity factor becomes MF = $1.5\sqrt{10/250} = 0.30$, so $D = 0.30 \times 442.40 = 132.72$, the add-on falls to $0.50\% \times 132.72 = \$0.664$ million, and

$$\text{EAD} = 1.4 \times 0.664 = \$0.93 \text{ million}$$

A 70% reduction in the add-on, purely from margining. This is the framework's central incentive and it is deliberate: the margin period of risk is the only window in which an unhedged move can hurt, so shortening it shortens the exposure. Chapters 2 and 8 described clearing and margin as regulatory mandates; this example is the economic reason a dealer would want them anyway. Capital, not virtue, is what moved the market.

Example 27.2's 70% reduction came from the maturity factor alone, and (27.7) contains a limit that is not obvious. The margined factor is capped at the unmargined value of one, and setting $1.5\sqrt{\text{MPOR}/250} = 1$ gives $\text{MPOR} = 250/1.5^2 = 111$ business days: beyond about five months of close-out risk, margining provides no add-on relief at all. The formula's logic is that exposure grows with the square root of time, so a shorter close-out window means a smaller possible move. But past 111 days the window is as long as the unmargined horizon and there is nothing left to shorten.

The 1.5 is a ratio of two averages. The supervisory factors are calibrated to an unmargined exposure averaged over one year, and an exposure growing as $\sqrt{t}$ averages to two-thirds of its one-year value, since $\int_0^1 \sqrt{t}\, dt = 2/3$; a margined exposure is the same ten-day move at every date, so it averages to the full $\sqrt{\text{MPOR}/250}$. Dividing by $2/3$ puts the two on one scale, and $1/(2/3) = 1.5$.

The practical content is in the middle of the curve. A position that genuinely takes thirty days to unwind (a concentrated equity book, an illiquid credit portfolio) keeps 52% of the unmargined add-on even under a daily margin agreement. The right panel of Figure 27.2 shows where the relief comes from and how quickly it runs out.

And that is the assumption Section 27.14 breaks. The regulatory ten days is a claim about liquidity, not the calendar; a dealer that applies it to a position worth many days of trading volume has used an input it did not verify, which is the failure mode of Chapter 21 arriving in a capital formula rather than a pricing model.

27.4 CAPITAL UNDER THREE REGIMES

EAD is not capital. Multiplied by the counterparty's risk weight RW it gives **risk-weighted assets** (RWA), and those times the required capital ratio CR give the capital the trade consumes:

$$\text{RWA} = \text{EAD} \times \text{RW}, \qquad \text{capital} = \text{RWA} \times \text{CR} = \text{EAD} \times \text{RW} \times \text{CR} \tag{27.11}$$

Neither factor is the bank's to choose. The supervisory risk weights are prescribed by counterparty type:

- 0% for a high-grade sovereign;
- 2% for a trade against a **qualifying CCP** (one licensed and supervised under rules consistent with the international standards for clearing houses);
- 20% for a highly rated bank;
- 100% for a typical unrated corporate;

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- more for a speculative-grade one.

CR is the minimum ratio Section 1.12 described, 8% plus the buffers a jurisdiction imposes. A bank with model approval estimates some risk weights itself, within supervisory floors, which is the internal-ratings route of Section 39.10.

Put the steps together and the framework's effect on market structure becomes a single table.

Example 27.3 Why everything moved to clearing

Example 27.2's swap at zero mark-to-market, against a corporate at a 100% risk weight, with a 10.5% capital requirement (the 8% minimum plus the 2.5% capital conservation buffer, a layer a bank may draw down under stress only at the price of limits on its dividends). The EAD is the $3.10 million computed there, carried to a third decimal so that the capital column can be checked.

	EAD	Risk wt.	RWA	Capital
Uncleared, unmargined	$3.097m	100%	$3.097m	$0.3252m
Uncleared, margined (10d)	$0.929m	100%	$0.929m	$0.0975m
Cleared at a qualifying CCP (5d)	$0.657m	**2%**	$0.013m	$0.00138m

The ratio is 236 : 71 : 1: clearing costs 0.4% of the capital of leaving the trade uncleared and unmargined.

Two changes drive it, and they compound. Clearing shortens the margin period of risk to five days, so MF = $1.5\sqrt{5/250} = 0.21$ and EAD falls 79% to $0.657m; the 2% weight for trade exposures against a qualifying CCP is then a further factor of fifty.

Turning capital into a running charge at a 10% hurdle rate (Section 9.7's KVA calculation) gives the number a client actually sees:

	bp per year of notional
Uncleared, unmargined	3.25
Uncleared, margined	0.98
Cleared	0.01

Three basis points a year sounds small, and on a swap that trades at a fraction of a basis point of bid–ask it is decisive. A dealer quoting the same swap to a cleared and an uncleared client is quoting from different cost bases, and the gap is larger than the spread.

This is the answer to a question Chapter 2 raised and did not settle. The clearing mandate covered standardized products; it did not force the rest of the market to clear voluntarily, and much of it did anyway. *The table is why.* Capital rules did more to move the market than the mandate did.

The table omits one qualification. A clearing member also holds capital against its default fund contribution, which the 2% line above excludes. The charge is set for the member, not the trade: it is the member's share, in proportion to its contribution, of the clearing house's own hypothetical capital requirement, computed as though the clearing house were a bank, and it can be no lower than that of a 2% risk weight on the contribution. For an illustrative $100 million contribution the floor alone is $2\% \times 100 \times 10.5\% = \$0.21$ million; spread over 10,000 cleared swaps like this one it adds $21 to each, against the $96,000 that clearing saves on each margined uncleared swap. The risk-sensitive charge is larger than its floor, but it would have to be thousands of times larger to change the ordering. It is the price of the mutualization of Chapter 2, a charge against a loss caused by somebody else's counterparty.

27.5 THE MEANING OF $\alpha = 1.4$

The multiplier in (27.1) is the most misread number in the framework. It looks like a measurement—exposures are 40% larger than they appear—and it is not one.

Under the internal model method, a bank computes an exposure profile by simulation, reduces it to **effective expected positive exposure** (the running maximum of expected exposure,

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averaged over the first year) and treats the result as a loan-equivalent amount. But a loan equivalent computed this way understates the capital a full portfolio credit model would require, because it ignores the correlation between exposures and defaults across counterparties and the granularity of the portfolio: a book concentrated in a few large counterparties is less diversified than the loan formula assumes.

α is the supervisory bridge over that gap. Its 1.4 is a calibration from studies of representative portfolios, with a floor of 1.2 for banks permitted to estimate their own.

So the 40% uplift is a buffer for model error, not a property of any trade; reading it as a real 40% higher exposure leads a desk to try to hedge something that does not exist.

A bank with supervisory approval keeps the internal model method because it simulates the actual netting set, margin terms and collateral, where SA-CCR applies prescribed factors. The benefit has a limit. Under the output floor, a bank on the internal model method computes counterparty exposure under SA-CCR as well, for the standardized base of the floor, so its total risk-weighted assets cannot fall below 72.5% of what that base gives.

27.6 THE CAPITAL CHARGE FOR CVA

A separate charge covers the risk that CVA itself moves: that a counterparty's credit spread widens, marking down the value of the derivative book, without anyone defaulting. The Basel Committee's 2011 announcement of the charge put roughly two-thirds of the crisis's counterparty credit losses in these mark-to-market movements, which the earlier framework had not capitalized. Section 9.7 explained why the charge exists and why an action that lowers the CVA price need not lower it.

The framework offers two approaches, and a simpler option for small banks. The **basic approach** (BA-CVA) is the default: a formula on exposures and on supervisory risk weights set by the counterparty's sector and rating. A bank whose non-centrally-cleared derivatives fall below a notional of €100 billion may instead set the charge equal to its counterparty credit capital. The **standardized approach** for CVA (SA-CVA) is a sensitivity-based calculation, delta and vega against credit spreads and the market factors driving the exposure, and is closer to how a desk actually hedges; it requires supervisory approval. It is a different standardized approach from Section 27.3's, which measures the exposure this charge sits on.

27.7 HEDGING CVA, AND THE EXEMPTIONS

The standardized CVA charge is sensitivity-based, which means eligible hedges reduce it. Which hedges are eligible, and by how much, turns out to determine what the desk of Section 9.10 actually trades.

A single-name CDS on the counterparty is recognized almost in full under the standardized approach, because the charge is computed on that counterparty's spread and the CDS is written on exactly it: buy the protection and the charge falls by what the protection removes. The basic approach caps the relief: however complete the hedge, it removes at most three-quarters of the charge.

An index CDS is recognized in part. The framework imposes a correlation between the index and the issuer, so some of the hedge's benefit is disallowed however well it works. A proxy (a CDS on a different issuer in the same sector and region) is recognized less still. This matters because for most counterparties there is no single-name CDS at all. Chapter 25's case study of the market that shrank established that liquid single-name CDS now exists for a few hundred issuers, while a dealer has thousands of counterparties.

So the instrument a desk can actually buy is the one the framework recognizes least. The consequence is a genuine dilemma rather than a rule to memorize: an index hedge removes most of the economic risk, earns partial capital relief, and introduces a basis (the index-to-single-name

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gap of Section 25.6) that the CVA charge still capitalizes, because the index is recognized only in part, and that shows up in earnings every quarter. Hedging the risk and hedging the capital charge are different trades, and the desk that does both holds a position it wanted neither of.

Then there are the exemptions, and they are large. The European framework exempts transactions with non-financial counterparties, sovereigns, and pension funds from the CVA capital charge, on the argument that these clients cannot post collateral and would be priced out of hedging. The United States and Canada did not adopt it. There the relief comes only from the framework's own carve-out, under which a trade cleared through a qualifying CCP is outside the CVA charge altogether, so what matters is how a trade is cleared rather than who is on the other side of it.

Whatever one thinks of the policy, the mechanical consequence should be stated. *The exposure is unchanged; only the capital is.* A European dealer's CVA on its corporate book is a real, volatile, unhedged number producing earnings swings and no capital requirement—and a dealer that *does* hedge it takes on market risk that is capitalized, while the CVA capital charge it is hedging against was already zero and cannot fall further.

The rule therefore discourages hedging the very exposures it chose not to charge for. That is an unusually clean example of the problem this chapter keeps returning to: a capital framework does not merely measure behavior, it selects it, and leniency in one place reappears as distortion in another.

27.8 DVA, AND WHY CAPITAL REFUSES IT

Section 9.3 established that a firm's own default risk produces a debit valuation adjustment (a gain when the firm's own credit deteriorates), that accounting standards require it to be recognized, and why regulators refuse it: a gain that can be monetized only by defaulting is not a buffer against default. The rule is that DVA gains and losses are filtered out of **common equity tier 1**, the highest-quality layer of capital (ordinary shares and retained earnings), so a bank whose spread widens may book a large accounting profit of which none counts as capital. Fair value and capital are answering different questions about the same liability, and the same reasoning is why Section 27.6's capital charge is computed on CVA gross of DVA while the accounting number is net.

27.9 WRONG-WAY RISK

Section 9.9 defined wrong-way risk (exposure that grows precisely where the counterparty is least able to pay) and showed that it makes the exposure conditional on default, not the average, the amount CVA must insure. Section 8.7 deferred the general treatment to here, and Example 27.4 puts a size on that substitution for CVA.

Example 27.4 What the correlation costs

A counterparty with a 3% annual default probability, 60% loss given default, and an average exposure of \$10 million has, under independence, CVA $= 0.60 \times 0.03 \times 10 =$ \$0.18 million. Putting $E[\text{exposure} \mid \text{default}]$ in place of the \$10 million multiplies that by the ratio of the two, so a correlation strong enough to double the conditional exposure doubles the charge, to \$0.36 million. The multiplier is linear in the conditional exposure ratio, and it is not small.

The calculation contains no correlation parameter. The conditional exposure ratio is the only thing that matters, and it is one number rather than a joint distribution, which is convenient, and also the reason the quantity is so hard to estimate. It is a conditional expectation on an event that has never occurred for this counterparty.

Wrong-way risk divides in two, and the distinction determines what can be done about it.

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Specific wrong-way risk is structural, like Section 9.9's sovereign case: taking a counterparty's own shares as collateral, or entering a commodity swap with a producer whose solvency depends on that commodity's price. These are identifiable in advance from the trade's economics. Where the counterparty is legally connected to the underlying or reference entity, as it is to its own shares, the regulatory response is to require the exposure to be computed as if that entity had already defaulted: a jump to default, with no correlation estimated. It is a rule rather than a model, because the model would need a correlation no one can estimate.

General wrong-way risk is milder: the counterparty's credit is correlated with the market factors driving the exposure. It is handled through the α of Section 27.5, whose calibration allows for that correlation, and by calibrating the exposure model to a period of stress.

The recurring lesson from three chapters generalizes. Section 24.4's recovery-default correlation, Section 26.4's excess spread that vanishes with losses, and this section's exposure-default correlation are the same failure: a protection whose value is negatively correlated with the need for it. Once named, it is checkable: for any mitigant, ask what it is worth in the state that makes it necessary.

27.10 THE LEVERAGE RATIO AS A BACKSTOP

Everything so far is risk-based: exposure is scaled by a risk weight, so a safer counterparty costs less capital. Alongside it sits a deliberately risk-*insensitive* constraint. The **leverage ratio** is tier 1 capital (common equity tier 1 plus the perpetual instruments that rank just behind it) divided by total exposure (assets, derivative exposure, securities financing and off-balance-sheet items, none risk-weighted), and it must clear a minimum LR, so every trade costs capital whatever its quality.

Two constraints on the same trade means one of them binds, and which one is arithmetic. Both sides are capital held against the trade. The leverage rule asks for $\text{EAD} \times \text{LR}$, since the trade adds its EAD to total exposure whatever its quality; the risk-based rule asks for the $\text{EAD} \times \text{RW} \times \text{CR}$ of (27.11). A bank must satisfy both ratios at once, and the same capital counts toward each, so a trade costs the larger of the two and never their sum. The first is larger when

$$\text{EAD} \times \text{LR} \;>\; \text{EAD} \times \text{RW} \times \text{CR} \qquad \Longleftrightarrow \qquad \text{RW} \;<\; \text{LR}/\text{CR}$$

so with a 10.5% capital ratio:

Leverage requirement	Leverage binds below a risk weight of
3% (baseline)	29%
5% (largest US banks, before April 2026)	48%
6%	57%

Sovereigns at 0%, qualifying CCPs at 2%, highly rated banks at 20%: all of the highest-quality exposures fall below the crossover. For exactly those businesses, capital is set by a measure that does not care that the risk is low.

The crossover is only a first approximation, and both refinements move it the leverage ratio's way. The leverage exposure measure uses its own version of EAD, recognizing collateral more grudgingly than (27.10) does, so a well-collateralized book looks larger to it than to the risk-based rule. And the two ratios are measured against different capital: tier 1 for leverage, total capital for the 10.5%, and since tier 1 is the narrower and costlier layer, a point of leverage requirement weighs more than a point of the risk-based one.

This is intended. A backstop that varied with risk weights would inherit whatever error is in them, and Chapter 26's pre-crisis experience is a story about risk weights being wrong. The leverage ratio bounds what the risk-based framework can be talked into.

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But the cost is specific and has been observed. The businesses it makes expensive (Treasury market-making, repo intermediation, client clearing) are low-margin, high-volume, and low-risk. A constraint that is insensitive to risk pushes a dealer out of its safest activities first, because those are where the constraint bites hardest relative to the revenue.

In March 2020 this became visible. Treasury market functioning deteriorated under a surge in volume, dealer balance sheets were constrained, and the supervisory response included temporarily excluding Treasuries and central bank reserves from the leverage exposure measure. The exemption expired in March 2021 and was not renewed; the recalibration of the US enhanced leverage ratio, finalized in November 2025 and effective April 1, 2026, lowered the requirement on all exposures instead, replacing the largest banks' fixed 2% buffer with half of each bank's global systemically important bank surcharge. The underlying question, whether a backstop should be relaxed at exactly the moment the activity it constrains becomes systemically important, has no comfortable answer.

Chapters 6 and 8 both rest on dealers intermediating at low margin, so a rule that taxes intermediation by volume rather than risk determines market liquidity and should be read as doing so.

27.11 HOW A BANK'S CAPITAL REQUIREMENT IS BUILT

Everything so far has priced one trade's capital through (27.11). A bank's requirement is assembled from many such pieces, and seeing the whole explains both what a derivative's capital competes with and which lever a desk can actually pull. The assembly has a numerator, the capital the bank holds, and a denominator, the risk-weighted assets it holds it against.

The numerator. Capital comes in layers, ranked by how readily they absorb loss, and each layer must be at least a set share of RWA. These are the **Pillar 1** minimums:

- *Common equity tier 1* (CET1), the ordinary shares and retained earnings of Section 27.8: at least 4.5%.
- *Tier 1*, which adds perpetual instruments that can be written down or converted: at least 6%.
- *Total capital*, which adds subordinated debt: at least 8%.

On top sit buffers, all of them met with CET1:

- the **capital conservation buffer**, 2.5%;
- the **countercyclical buffer**, 0 to 2.5%, which each jurisdiction sets according to its credit cycle;
- for a **global systemically important bank** (G-SIB), a surcharge of 1% to 3.5%, set by the bank's size, interconnection and complexity.

A bank that dips into the buffers is not in breach, but its dividends and bonuses are restricted until it rebuilds them. In the United States the conservation buffer of the largest banks is replaced by the stress capital buffer of Section 39.12, which is at least as large. Two further layers complete the framework:

- **Pillar 2**, the supervisor's bank-specific add-on for risks the formulas miss, such as concentration and interest rate risk in the banking book;
- **Pillar 3**, the public disclosure that lets the market check the arithmetic.

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The denominator. Total RWA is the sum of one term per risk type, each computed by its own rule:

$$\text{RWA} = \text{RWA}_{\text{credit}} + \text{RWA}_{\text{CCR}} + \text{RWA}_{\text{CVA}} + \text{RWA}_{\text{market}} + \text{RWA}_{\text{op}} + \text{RWA}_{\text{sec}} \tag{27.12}$$

- *Credit*: loans and bonds in the banking book, weighted by Table 27.1 below or by the internal-ratings formula of Section 39.10. It is usually by far the largest term.
- *Counterparty credit*: this chapter's SA-CCR exposure times the counterparty's risk weight.
- *CVA*: the charge of Section 27.6, times 12.5.
- *Market*: the trading-book charge of Chapter 39, times 12.5.
- *Operational*: losses from failed processes, people and systems, fraud and litigation among them. The charge is a standardized formula on a **business indicator** built from the bank's income, multiplied by coefficients of 12% to 18% that rise with size and, where the supervisor allows, scaled by the bank's own loss history; times 12.5.
- *Securitization*: a framework of its own, sketched below.

The factor 12.5 is the reciprocal of 8%: it turns a capital charge into the RWA that would require it. Under the output floor of Section 39.10, total RWA from internal models may not fall below 72.5% of what the standardized rules give for every term together.

Example 27.5 From risk-weighted assets to a capital requirement

A G-SIB in the 1.5% surcharge bucket, in a jurisdiction with a 0.5% countercyclical buffer, reports RWA in billions of dollars:

Risk type	RWA ($bn)
Credit (banking book)	600
Counterparty credit (SA-CCR)	40
CVA	30
Market (Chapter 39)	60
Operational	100
Securitization	20
Total	**850**

The buffers add 2.5% + 1.5% + 0.5% = 4.5% of CET1 to every layer, so the requirements are

	Ratio	Capital ($bn)
CET1	4.5% + 4.5% = 9.0%	76.50
Tier 1	6.0% + 4.5% = 10.5%	89.25
Total capital	8.0% + 4.5% = 12.5%	106.25

before any Pillar 2 add-on. The bank's leverage exposure is $2,500 billion, and a G-SIB's leverage requirement is the 3% minimum plus half its surcharge, 3% + 0.75% = 3.75%, or $93.75 billion of tier 1. That exceeds the risk-based tier 1 figure of $89.25 billion, so the leverage ratio binds. The reason is Section 27.10's crossover: the bank's average risk weight is 850/2,500 = 34%, below the 3.75%/10.5% = 35.7% at which the two rules cost the same.

Read the table for what it says about derivatives. Counterparty and CVA risk together are 70 of 850, about 8%, and the market term another 7%: the capital a derivatives business consumes is a minority of the bank's, which is why its marginal cost depends on what the rest of the balance sheet is doing. Here, with leverage binding, a desk that lowered its risk weights would save nothing, and one that shrank its exposure would.

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Standardized credit risk weights. A bank without model approval, and every bank for the output floor, weights its banking-book exposures by Table 27.1. The weight depends on the type of borrower and, in the Basel text, on its external rating.

Table 27.1: Standardized risk weights for credit exposures in the Basel framework (%), by external rating. Jurisdictions that do not use ratings, the United States among them, substitute their own schedules.

	AAA to AA	A	BBB	BB	B	Below B	Unrated
Sovereign	0	20	50	100	100	150	100
Bank	20	30	50	100	100	150	40–150
Corporate	20	50	75	100	150	150	100

Regulatory retail 75; residential mortgage 20 to 70 by loan-to-value; equity 250; subordinated debt 150

An off-balance-sheet item enters through a **credit conversion factor**, the share of it treated as drawn:

- 100% for a guarantee that substitutes for a loan;
- 40% for an undrawn commitment;
- 10% for a commitment the bank can cancel without notice;
- 20% for a short-term trade letter of credit.

A \$200 million undrawn revolving facility to a BBB corporate therefore carries $200 \times 0.40 \times 0.75 =$ \$60 million of RWA before a cent is lent.

Credit risk mitigation. The weights apply to the exposure the bank actually bears, and three things reduce it:

- *Collateral* reduces the exposure itself. Under the comprehensive approach the exposure after mitigation is $E^* = \max\{0,\ E(1+H_e) - C(1-H_c-H_{fx})\}$, where the supervisory haircuts H grow with the collateral's volatility (zero for cash, a few percent for a short government bond, 20% or more for equities), with a further 8% if the collateral is in another currency.
- *Guarantees and credit derivatives* work by substitution: the protected part of the exposure takes the risk weight of the protection seller instead of the borrower's.
- *Netting*, the subject of Section 27.2, reduces exposure for derivatives and securities financing.

Substitution is how a bank's purchase of CDS protection (one of the uses Chapter 25 lists) turns into capital relief (Example 27.6), and it comes with conditions:

- the protection must be direct, irrevocable and unconditional;
- if the CDS does not count restructuring as a credit event, only 60% of it is recognized, because the loss a restructuring imposes falls on the lender unprotected;
- if the protection runs out before the loan, it is recognized only in proportion, as $P_a = P\,(t-0.25)/(T-0.25)$ for protection P with t years left against a loan with T, and not at all if it has three months or less to run.

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Example 27.6 What a CDS hedge does to a loan's capital

A bank has a $100 million five-year loan to an unrated corporate, weighted 100%, at the 10.5% requirement of Example 27.3. It buys five-year protection from a bank rated AA, weighted 20%.

	RWA ($m)	Capital ($m)
Unhedged	100.00	10.50
Hedged, CDS covers restructuring	20.00	2.10
Hedged, CDS without restructuring	52.00	5.46
Hedged with a three-year CDS	53.68	5.64

The full hedge moves the whole loan to the protection seller's 20%. Without restructuring, 60% of it is recognized: $60\times0.20+40\times1.00 = 52$. With a three-year contract against a five-year loan, the recognized amount is $100\times(3-0.25)/(5-0.25) = 57.89$, and $57.89\times0.20+42.11\times1.00 = 53.68$.

The full hedge saves $8.40 million of capital. At the 10% hurdle rate of Example 27.3 that is worth $0.84 million a year, 84 bp of the loan. So for a borrower whose CDS trades below 84 bp the capital saved covers the hedge's cost on its own, and for one above it the hedge is bought for the risk it removes, with the capital relief covering only part of its cost. The comparison also leaves out two costs the hedge adds back: the CDS is itself a derivative, with a counterparty exposure under SA-CCR and a CVA charge, both small for a single bank-sold contract. A $100 million loan secured by $60 million of cash, by contrast, needs no counterparty at all: $E^* = 100 - 60 = 40$, and RWA fall to $40 million.

Securitization. A securitization position has its own risk-weight function, because a tranche's risk depends on where it sits in the waterfall of Section 26.2 as much as on the pool. The framework computes the capital the pool would need if held whole, K, and charges each tranche according to how much of that it absorbs:

- a tranche below K is weighted 1,250%, the reciprocal of 8%, so it costs capital dollar for dollar;
- a tranche above it is charged by a formula that falls with the attachment point, down to a floor of 15% for a senior tranche (10% for one meeting the simple, transparent and comparable criteria).

Those two numbers explain the significant risk transfer trades of Chapter 26. A bank that sells the first-loss and mezzanine tranches of a loan book, and keeps the senior one at or near the floor, replaces the loans' risk weights with a fraction of them. The regulator's test for whether the transfer is real is exactly how much risk has left the bank.

The pieces fit one way. A desk's capital is a share of a requirement whose binding constraint (the risk-based ratios, the leverage ratio, the output floor, or a stress test) is set by the whole bank. Which one binds decides which lever moves capital: risk weights matter only while a risk-weighted rule binds, exposure only while the leverage ratio does, and neither while the floor does.

27.12 CLIENT CLEARING AND THE COST OF BEING AN INTERMEDIARY

Example 27.3 showed the capital benefit of clearing to the party whose trade is cleared. Somebody has to provide the access, and that role's economics are worse than the table shows.

A client does not face the clearing house directly. *A clearing member stands between them*, facing the CCP on one side and the client on the other, and the two legs do not cancel: the member owes the CCP whatever the client fails to pay, and holds the client's margin against it.

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Three capital consequences follow, and together they explain a surprising market structure.

The member's exposure to the client is a derivative exposure and capitalized as one, even though the member has no market risk on it: each leg's market moves offset the other's, but the two face different counterparties and so cannot be netted.

Client margin is segregated, so the member cannot use it to fund. Chapter 8's segregation requirement, which exists so that the client's collateral survives the member's failure, means the member funds its own contribution to the CCP without the benefit of what it holds. That is Section 9.5's MVA arriving as an operating cost rather than as a charge on a trade.

And the leverage ratio counts the exposure. Section 27.10's crossover applies with force here: the risk weight against a qualifying CCP is 2%, so the risk-based charge is negligible and the leverage requirement binds. Whether segregated client initial margin may be recognized as reducing the leverage exposure measure has been changed more than once and remains among the most consequential technical questions in the framework, because it determines whether client clearing is a viable business.

The observable result is concentration. Client clearing in the major currencies is provided by a small number of banks, and several large institutions have exited. A framework designed to reduce concentration of counterparty risk has concentrated access to the mechanism that reduces it—a reminder that capital rules allocate activity as well as loss-absorbency.

And it interacts with Chapter 2's central concern. Porting a defaulted member's client positions requires another member able to take them, so the fewer members there are, the less credible porting becomes—and porting is what the whole client-protection story rests on.

27.13 CONCENTRATION AND THE LIMITS OF BILATERAL MEASUREMENT

Everything above measures a dealer's exposure to a counterparty. Nothing above measures a counterparty's exposure to the market, and that gap is the chapter's most important weakness.

The framework is bilateral by construction: each firm computes its own exposure to each of its counterparties, using data it has. A client's positions at other dealers are not data it has.

So a counterparty that is prudently margined at every dealer individually can be catastrophically levered in aggregate, and no participant in the system is in a position to observe it. The only parties who could are the client, who has no obligation to disclose, and the regulators, whose reporting captures cleared and reported trades but not, in useful time, a consolidated view of one client's leverage across brokers.

In March 2021 that gap cost the banks that had measured their own exposures correctly more than $10 billion.

27.14 ARCHEGOS

Case Study 27.1 What total return swaps made possible

In March 2021 a family office defaulted on positions built through total return swaps at several prime brokers. Losses to the banks totaled over $10 billion, with a single institution losing more than $5 billion—one of the largest single-counterparty losses in the derivatives market's history, and it involved no exotic instrument and no pricing error.

The arithmetic is simple enough to be alarming.

Take a fund with $10 billion of capital taking five times leverage through total return swaps, split across six prime brokers:

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Total gross exposure	\$50bn
What each broker sees	\$8.3bn
Initial margin held by each, at 15%	\$1.25bn

The 15% is a prime brokerage number for a diversified equity book. Now the underlying positions, concentrated in a small number of issuers, fall:

Move	Loss on gross	
−10%	\$5.0bn	survivable
−20%	\$10.0bn	client's capital exhausted
−30%	\$15.0bn	client insolvent; loss beyond the margin falls on the brokers

The client's capital is exhausted at a 20% move, which for an equity portfolio concentrated in a few stocks is an ordinary week. At 30%, each broker holds \$1.25 billion of margin against a \$2.5 billion loss—*a shortfall of \$1.25 billion each*. This assumes a broker can look only to the margin it holds, since the client's other \$2.5 billion of capital is not posted. Were all of it collected, the total shortfall would be \$5 billion rather than \$7.5 billion.

Four failures stand out, and they are worth separating because only one is about the instrument.

Invisibility. Section 7.12 said total return swaps are how leverage accumulates quietly, and this is what that meant. Each broker's exposure was individually unremarkable. The swap format also kept the positions out of ownership disclosures, so the concentration was not visible to the market either.

Margin that was not risk-based. Initial margin was set at levels reflecting competition for the client's business rather than the concentration and liquidity of the underlying. The margin period of risk assumption of Example 27.2 (that a portfolio can be closed out in ten days) was false for a position worth many days of trading volume in each issuer.

Concentration measured against the wrong thing. An \$8.3 billion exposure is modest against a large bank's balance sheet and enormous against the free float of a mid-cap stock. The relevant denominator is liquidity, not capital, which is the same lesson Section 23.12's single-stock snapshot drew from the other direction.

And a liquidation that moved the price. When the brokers sold, they sold the same stocks into each other, which is Chapter 22's crowding and Chapter 23's feedback in the least sophisticated form: six firms with the same collateral trying to sell it in the same week. The brokers that sold first lost least, which converted a credit problem into a race.

What the framework did and did not catch. SA-CCR would have registered each broker's exposure and required capital against it; no bilateral framework registers that five other institutions hold the same exposure to the same client. The measurement was correct and the aggregate invisible, which is the chapter's conclusion.

Supervisory responses have focused on margin practices for concentrated positions, prime brokerage risk governance, and disclosure of large synthetic equity positions. None fully solves the aggregation problem, which needs either a consolidated view no institution has or a disclosure obligation on the client.

27.15 INITIAL MARGIN AS THE REAL ANSWER

Chapter 8 described the uncleared margin rules as documentation and operations. What they buy is capital, and the two margin types buy different parts of it.

Variation margin removes current exposure: it is settled daily and brings the mark-to-market back to zero, which is the RC of (27.1). *Initial margin removes the gap risk*: the move between

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the last margin call and the close-out, which is precisely the margin period of risk that sets the maturity factor in Example 27.2.

The two differ in kind. Variation margin is a transfer of value that has already happened; initial margin is a buffer against value that has not, which is why it must be segregated and cannot be rehypothecated.

Its cost is Section 9.5's MVA, and its benefit is the EAD reduction of Example 27.2. A dealer's decision about how much initial margin to demand is therefore a trade between a funding cost it pays and a capital cost it avoids, and Archegos is what happens when competition pushes that trade toward the client.

The structural point is that initial margin is the only mitigant that works without information. Netting requires legal certainty; wrong-way risk requires identifying the correlation; concentration limits require knowing the aggregate. Initial margin protects against a move of a given size whatever its cause, which is what makes it right when what you cannot see is what hurts you.

27.16 WHAT THE FRAMEWORK DOES NOT CAPTURE

The framework leaves four things unmeasured, and each has cost someone money.

Aggregate leverage. Section 27.13's gap, which nothing bilateral sees.

Liquidity of the underlying. SA-CCR's supervisory factors are by asset class, not by the liquidity of the specific position. A concentrated position in an illiquid issuer and a diversified one of the same notional carry the same add-on.

Correlated liquidation. The framework asks what a firm loses if one counterparty fails. It does not ask what it loses if that failure forces five other firms to sell the same assets.

And the model risk in the exposure itself. The exposure profile is a simulation output, and Chapter 21 established that model choice moves such outputs materially; SA-CCR's standardization is partly a response, at the cost of being wrong in a known direction outside its calibration.

None of this is an argument against the framework. The framework substantially improved on what preceded it, and the CVA charge in particular addressed a loss that had gone entirely uncapitalized. The argument is that a capital framework measures what a firm can see, and the losses that matter most have repeatedly come from what the whole system could not.

Capital is a buffer, not a control. It fixes what must be held once a position exists; it does not decide whether to take the position. Every episode in this chapter turned on a judgment made inside a firm (how much initial margin to demand of a client, which limit to override, how fast a concentrated holding could be sold), and no calibration reaches those. Regulation can make imprudence expensive, which is most of what the post-crisis framework achieved; declining to be imprudent is left to the institution, and Section 27.14 is what it costs when a firm does not.

27.17 WHO BEARS WHAT

Dealers bear counterparty risk, capitalize it under this chapter, price it under Chapter 9, and hedge it in the credit market of Chapter 25: three activities with three different numbers, which Section 9.11 warned would not reconcile.

Clearing houses bear it mutualized, under the default waterfall of Chapter 2. Moving a trade to a CCP converts a bilateral exposure into a contribution to a shared pool, which is a real reduction in system-wide exposure and a real concentration of it in one place.

Clients bear the cost, through the charges Chapter 9 enumerated, seen or not.

And the system bears the residual, which is the part this chapter says cannot be measured bilaterally. *Whoever bears that is decided after the fact*, and in 2008 and 2021 it favored whoever sold first.

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SUMMARY

CVA is a price and regulatory capital is a buffer, computed by different rules for different purposes. A derivative's exposure is an option on the market, bilateral, belonging to the whole netting set, and dependent on mitigants that can fail. Netting is the largest reducer of it: a portfolio whose trades largely offset can carry a net exposure that is a small fraction of its gross, a reduction obtained from a legal clause, so a jurisdiction's netting opinion is worth more to a dealer than any risk model.

SA-CCR sets EAD = $1.4 \times$ (RC + PFE), with RC the collateralized mark-to-market (or the exposure a margin agreement's threshold leaves uncalled) and PFE built from add-ons by asset class. Daily margin cuts a swap's exposure sharply through the maturity factor alone, which scales the add-on to the margin period of risk rather than to the trade's maturity, yet a position that takes weeks to unwind keeps much of its add-on. The capital a swap consumes falls by a large multiple from uncleared to margined and again from margined to cleared, which is why capital rules moved more of the market to clearing than the mandate did. The α of 1.4 is a buffer for model error, not a property of any trade.

A separate CVA charge covers spreads widening without default, two-thirds of the crisis's counterparty losses and uncapitalized before it; hedge recognition is uneven, and Europe's exemptions discourage hedging exactly the exposures the charge was built for. Capital filters DVA out of common equity tier 1 because a gain realizable only in default is not a buffer. Wrong-way risk, recovery–default correlation, and vanishing excess spread are one failure mode: protection whose value falls as the need for it rises.

The leverage ratio binds wherever the risk weight is below LR/CR, so it makes the safest, thinnest businesses expensive first: Treasury market-making, repo, and client clearing. A bank's whole requirement stacks minimums of 4.5%, 6% and 8% of risk-weighted assets for common equity, tier 1 and total capital, then conservation, countercyclical and systemic buffers. Standardized credit weights rise as the rating falls; collateral reduces the exposure, and a guarantee or CDS substitutes the protection seller's weight, recognized only in part if the contract omits restructuring or ends before the loan. Which rule binds for the whole bank (risk-based, leverage, floor or stress test) decides whether a desk saves capital by cutting risk weights, by cutting exposure, or not at all. And the framework is bilateral: in 2021 a family office that looked prudently margined at each of its prime brokers, and was heavily levered across them through total return swaps, cost them over $10 billion with no exotic instrument and no pricing error. The measurement was correct and the aggregate was invisible. Initial margin is the one mitigant that works without information, and a capital framework measures only what a firm can see.

FURTHER READING

The consolidated Basel Framework, published online by the Bank for International Settlements, is the primary source for Section 27.11: its credit risk chapters hold the standardized weights, the conversion factors and the mitigation rules, and the Basel Committee's *Basel III: Finalising Post-Crisis Reforms* (2017) is the reform that set them. The Basel standards on SA-CCR and on CVA risk are the primary sources and are specific enough to compute from; anyone implementing either should read them rather than a summary, since the hedging-set definitions carry most of the practical content. Gregory's book on counterparty credit risk and XVA is the standard practitioner treatment and covers the pricing of Chapter 9 and the capital of this chapter in one place. On Archegos, the independent report commissioned by the largest loss-making bank sets out how risk limits were set and overridden; the supervisory reviews that followed say more about what changed than about what happened. On wrong-way risk, the literature is small, and the regulatory rule (compute as if default had occurred) remains more

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widely used than most of the models. On the aggregation gap of Section 27.13, the work on position transparency and on consolidated leverage reporting is active and unresolved.

PRACTICE QUESTIONS

27.1 Explain the difference between CVA as a price and CVA capital as a buffer, and why reducing one need not reduce the other.

27.2 Four trades with a counterparty mark at $+30, -12, +8, -20$. Compute the exposure with and without netting, and explain why a jurisdiction's netting opinion is described as worth more than any risk model.

27.3 Compute the SA-CCR exposure at default for a portfolio with a replacement cost of \$8 million and a PFE add-on of \$3 million. Then compute the maturity factor for a ten-day margin period of risk, state what it does to the add-on, and state what assumption the margin period embodies.

27.4 Compute the margin period of risk at which the maturity factor reaches one, and state what this means for an illiquid portfolio.

27.5 A bank pays fixed on a three-year \$100 million swap, unmargined, with no collateral. Compute its supervisory duration, effective notional and add-on. It then adds a seven-year receive-fixed swap of \$47.2 million, chosen so that the two have equal and opposite dollar duration. Compute the hedging set's effective notional and add-on, and explain why a position with no net rate sensitivity still carries most of the add-on, and what would change if the second trade had a four-year maturity instead.

27.6 Using Example 27.3, state the ratio of capital across the three regimes and the two changes that produce it, then convert a capital requirement of \$0.20 million on \$100 million of notional into a running charge in basis points at a 12% hurdle rate.

27.7 Explain why the clearing mandate is described as less important than the capital rules in moving the market to clearing.

27.8 Explain what $\alpha = 1.4$ is calibrated to bridge, and why it is not a statement about any individual trade.

27.9 Explain what the CVA capital charge covers that the default charge does not, why this loss went uncapitalized before the charge existed, and why the hedge a desk can actually buy is the one the framework recognizes least.

27.10 Explain why regulatory capital filters DVA out of common equity tier 1 while accounting requires it to be recognized.

27.11 A counterparty has a 4% default probability, 60% loss given default, and an average exposure of \$20 million. Compute CVA under independence, and again if the exposure conditional on default is twice the average. Distinguish specific from general wrong-way risk and state how each is treated.

27.12 Give three examples from this book of a protection whose value is negatively correlated with the need for it.

27.13 Derive the risk weight below which the leverage ratio binds at a 10.5% capital requirement and a 5% leverage requirement, and explain why a risk-insensitive constraint makes a dealer's safest businesses expensive first.

27.14 Explain why a clearing member's capital requirement is not eliminated by the fact that its client-facing and CCP-facing legs offset, and how the resulting concentration weakens the porting mechanism of Chapter 2.

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27.15 In the Archegos arithmetic, state the move at which the client's capital is exhausted and the shortfall per broker at a 30% move, taking margin as all a broker can collect, and explain why the relevant denominator for concentration is liquidity rather than the dealer's capital.

27.16 Explain why initial margin is described as the only mitigant that works without information, and state four things the capital framework does not capture.

27.17 In Example 27.5, the countercyclical buffer rises from 0.5% to 2.0%. Compute the new CET1, tier 1 and total capital requirements, and state whether the leverage ratio still binds and why the buffer change does not affect the leverage requirement.

27.18 A bank has a $50 million loan with four years to run to a BBB-rated corporate and hedges it with a two-year CDS, covering restructuring, bought from a bank rated A. Using Table 27.1, compute the loan's RWA before and after the hedge, and explain why protection shorter than the loan is recognized only in part.

27.19 Explain why a securitization tranche below the pool's own capital requirement is weighted 1,250%, and why the floor on the senior tranche's weight is what makes a significant risk transfer trade worth doing.

FURTHER QUESTIONS

27.20 Example 27.2 shows margining cutting the add-on by 70%. Derive the maturity factor's dependence on the margin period of risk, determine the period at which margining ceases to reduce the add-on, and discuss what this implies for a portfolio whose true close-out period is thirty days.

27.21 Section 27.9 argues that recovery-default correlation, disappearing excess spread, and exposure-default correlation are one failure mode. Formalize the common structure, propose a single diagnostic that would detect all three, and apply it to two mitigants used elsewhere in this book.

27.22 Section 27.13 identifies the aggregation gap. Design a reporting regime that would have revealed the Archegos position: specify what would be reported, by whom, to whom, and at what frequency; then state the costs (competitive, operational, and in terms of what the disclosure itself might cause).

27.23 The Archegos snapshot notes that brokers who sold first lost least. Model this as a game between the brokers, identify the equilibrium, and explain why a coordinated liquidation would have produced a better aggregate outcome and why it was not achievable. Relate your answer to Chapter 2's default waterfall, which is designed to solve exactly this problem for cleared trades.

27.24 Section 27.15 argues initial margin works without information. Construct the counter-argument that margin levels themselves require information about volatility, liquidity, and concentration, and reconcile the two positions by distinguishing what must be known to set the level from what must be known for the protection to function.

27.25 Section 27.16 says a capital framework measures what a firm can see. Compare this with Chapter 26's conclusion that a system loses the ability to see a risk before it loses the ability to bear it, and with Chapter 23's observation that 0DTE hedging demand is computable in advance. Use the three to argue either that observability is the binding constraint on financial stability or that it is not.

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